Diesel Engine
V 1163 TB 73L
Maintenance Schedule
M050449/02E

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- Ensure that sufficient fuel is available,
- Ensure that the combustion air is dry and clean,
- Use only dry, clean compressed air,
- Use only clean, filtered raw water.

Moreover, it is essential that:

- Maintenance tasks be completed by trained personnel,
- Suitable tools be used,
- Genuine spare parts and fluids and lubricants as per the current MTU Fluids and Lubricants Specification A001061 be used.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel; never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept and features the maintenance echelons W1 to W6 as outlined below.
Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

Main Features of Maintenance Tasks for MTU Engines.

Maintenance Echelon	W1:	Operational Checks.
	W2, W3 and W4:	Periodic maintenance tasks to be performed during out-of-service periods without need for engine disassembly.
	W5:	This echelon requires partial engine disassembly.
	W6:	Major overhaul. This echelon requires complete engine disassembly.

The time intervals according to which the W2 to W6 Maintenance Echelons and the relevant checks and tasks involved are to be completed are based on operational experience. They have been determined so as to ensure correct engine operation until the next scheduled maintenance echelon.
Use of a maintenance predictor can mean changes to the operating hours given in the Maintenance Frequency Chart.

Specific operational conditions may require modification of the Maintenance Schedule.

NOTE:
The codes on the following pages refer to the groups and subgroups of the Task Descriptions in Section G of the Engine Operation Manual.
This Maintenance Schedule may refer to parts which are not installed on your engine, these may be disregarded.

Application Group

1B: Fast vessels with high load factors

Load profile

Load referenced to fuel stop power	%:	100	85	15
Time element	%:	3	82	15

Maintenance Frequency Chart

Maintenance Echelon	W1:	Operational checks, daily	X
	W2:	Operating hours	500
		Limit, months	6
	W3:	Operating hours	1 000
		Limit, year	1
	W4:	Operating hours	4 000
		Limit, year(s)	2
	W5:	Operating hours	6 000 - 7 000
		Limit, year(s)	6
Intermediate check		Operating hours	10 000
		Limit, year(s)	8
	W6:	Operating hours	20 000
		Limit, year(s)	18

Maintenance Task Schedule -Overall System-

See page 4

Maintenance Task Schedule -Engine-

Code No.	One-time operations after the first 50 operating hours – With a new engine, after W5 maintenance echelon, or after W6 major overhaul	
G00.049	Attachments	Check tightness of securing screws and nuts
G06.101	Valve gear	Check valve clearances, adjust if necessary
G10.111	Intake air line	Remove protective grill (only with intake from atmosphere)
G12.311	Fuel pre-filter	Flush, replace filter insert
G12.321	Fuel duplex filter	Replace filter elements
G14.019	Engine coolant system	Clean strainer before coolant thermostat. Remove mesh
G16.002	Engine oil	Take sample and analyze
G19.011	Engine mounts	Check shock-buffer setting adjustments

Code No.	Maintenance Echelon W1: – Operational Checks	
G10.051	Exhaust system	Check exhaust gas color Drain condensate (if drain valve provided)
G10.181	Intake air line	Check condensate drain line for water discharge and obstructions
G12.311	Fuel pre-filter	Check differential pressure
G14.011	Engine coolant	Check level
G16.002	Engine oil	Check level
G84.001	Monitoring system	Carry out lamp test
G84.002	Engine operation	Check running noises Check engine and external pipework for leaks Check speeds, pressures and temperatures - if gauges are provided
G86.051	Compressed air system	Drain condensate